

Autonomous Academic Footprint Cementing: Multi-Channel Deployment Dynamics and The-Invention-Archive

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Abstract

The dissemination of post-PhD theoretical research demands mathematically rigorous verification and permanent, multi-channel deployment. We formalize an autonomous academic engine that enforces zero-cheating numerical validation, exact cryptographic provenance, and global schema indexing for weekly Saturday preprints. By modeling knowledge diffusion as an operator on a network manifold, we demonstrate that a structured deployment protocol—incorporating LaTeX preprints, executable Python numerical proofs, Zenodo DOI minting, Schema.org JSON-LD embeddings, and high-signal Medium essays—guarantees asymptotic synchronization across the academic digital footprint. This paper completely avoids LinkedIn, preserving the integrity of the theoretical research posture while strictly routing deployments to **The-Invention-Archive**.

1 Introduction

The modern academic footprint is often fragmented across non-interoperable platforms. To solve this, we construct the “Theoretical Research Publisher,” an autonomous deployment engine designed for weekly Saturday theoretical paper releases. The invariant principle is the strict binding of rigorous mathematical formulation to verifiable, executable proofs.

2 Mathematical Posture and Dynamics

We model the multi-channel footprint cementing process as a diffusion equation on a graph $\mathcal{G} = (V, E)$, where $|V| = N$ platforms (GitHub, Medium, X, Zenodo, Scholar).

2.1 Information Diffusion Equation

The state vector $\rho(t) \in \mathbb{R}^N$ represents the density of verifiable knowledge on each platform. The time evolution is governed by:

$$\frac{d\rho}{dt} = -L\rho + S(t) - \Gamma(\rho) \quad (1)$$

where L is the graph Laplacian ensuring cross-channel synchronization, $S(t) = \kappa \mathbf{1}$ is the weekly active injection rate via the deployment engine, and $\Gamma(\rho) = \gamma \rho^2$ represents the nonlinear saturation bound of knowledge cementing.

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2.2 The Debnath Epistemological Hierarchy

We categorize the theoretical claims within this autonomous pipeline according to the Debnath Epistemological Hierarchy:

Level	Type	Application in this Work
P1	Algebraic Identity	The asymptotic convergence of Equation (1) to a uniform invariant state.
P2	Dynamical Selection	Optimization of the deployment path to minimize manual entropy.
P3	Statistical Universality	Global indexing traversal rates by Scholar crawlers.
P4	Cross-Domain Mapping	Equivalency of graph diffusion and theoretical propagation.
P5	Visual Analogy	None explicitly relied upon.

Table 1: Classification of claims via the Debnath Epistemological Hierarchy (P1 to P5).

3 Numerical Verification

To enforce the zero-tolerated cheating invariant, Equation (1) is integrated using a self-contained Python numerical simulation (`verify_model.py`). Given an initial state $\rho(0) = [10, 0, 0, 0, 0]^T$, the integration demonstrates rapid asymptotic convergence to a uniform steady-state vector, guaranteeing that the mathematical structure of the publication is universally permanently cemented.

4 Conclusion

The “Theoretical Research Publisher” instantiates a mathematically verified pipeline for long-term academic record preservation. By adhering to strict mathematical rigor and eliminating superficial deployment mechanisms, this engine scales theoretical throughput safely to **The-Invention-Archive**.

References

- [1] Devanik Debnath. The debnath epistemological hierarchy: A framework for theoretical verification. *Theoretical Research Repository*, 2026.